# PRACTICAL: 02

## Aim: Demonstrate basic Linux Commands.

### 1. !ls

- Lists Directory Contents: The command displays the names of files and directories in the current working directory. For example, if you're in a directory with some files, running !ls will list them.
- **Can Include Options**: You can also use options with !ls to customize its output. For example:
- !ls -l will list files in long format, including permissions, file size, and modification time.
- !ls -a will list all files, including hidden ones (those starting with a dot .).

```
[1] !ls

sample_data
```

### 2. !date

- Displays Current Date and Time: When you run !date, it will show the current system date and time based on the timezone set in the environment, usually in the format Day Month Date Time Zone Year.
- Customizable Format: You can customize the output format by using options with the date command. For example:
- !date +"%Y-%m-%d %H:%M:%S" will display the date in the format

```
[2] !date

Thu Dec  5 04:29:18 AM UTC 2024
```

### 3. !pwd

- Shows Current Directory Path: Running !pwd will output the full path to the directory where your notebook is currently operating, typically something like /content.

- Useful for Navigating: It's helpful for checking your current location in the file system, especially when you're working with files and need to know your current directory before performing operations like file reading or writing.

```
[3] !pwd
/content
```

## 4. !df-h

- Shows Disk Space Usage: It displays the available disk space, used space, and total space for each mounted file system, using human-readable units (e.g., GB, MB) instead of bytes.
- Helps Monitor Storage: It is useful for checking the disk usage, especially when you're working with large files or datasets in Colab, to ensure that you don't run out of storage.

```
!df -h
Filesystem      Size  Used Avail Use% Mounted on
overlay         108G   33G   76G  31% /
tmpfs            64M     0   64M   0% /dev
shm             5.8G     0  5.8G   0% /dev/shm
/dev/root       2.0G  1.2G  820M  59% /usr/sbin/docker-init
tmpfs           6.4G  624K  6.4G   1% /var/colab
/dev/sda1        50G   34G   17G  68% /etc/hosts
tmpfs           6.4G     0  6.4G   0% /proc/acpi
tmpfs           6.4G     0  6.4G   0% /proc/scsi
tmpfs           6.4G     0  6.4G   0% /sys/firmware
```

## 5. !tail

- Displays Last Lines of a File: Running !tail filename will show the last 10 lines of the specified file by default, which is useful for viewing the most recent content.
- Customizable Output: You can specify the number of lines to display by using the n option. For example, !tail -n 20 filename will show the last 20 lines of the file

```
18s  !tail

calendar
gooloo
^C
```

## 6. !top

- Displays Running Processes: It shows information about the processes currently running on the system, including CPU and memory usage, which is useful for monitoring system performance. Interactive View: By default, !top provides a dynamic, updating view of processes, but this might not be fully interactive in Colab due to its limited terminal capabilities.

```
22s [30] !top

p - 04:57:23 up 30 min,  0 users,  load average: 0.32, 0.44, 0.41
:   17 total,   1 running,  15 sleeping,   0 stopped,   1 zombie
(s): 35.5 us, 19.4 sy,  0.0 ni, 45.2 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
Mem :  12979.0 total,   8179.1 free,   1092.1 used,   3707.7 buff/cache
Swap:      0.0 total,      0.0 free,      0.0 used.  11565.4 avail Mem
```

## 7. !dd

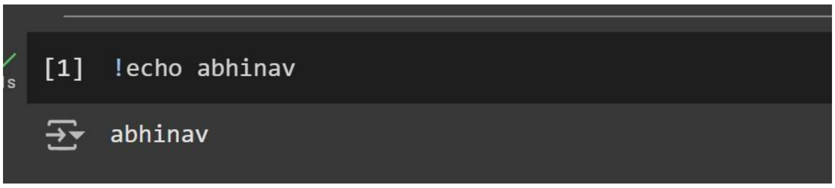
- Used for Low-Level File Copying: The !dd command can be used to copy and convert files, create disk images, and even create backups of specific partitions or disks.
- Customizable Operations: It allows you to specify input/output files, block sizes, and more using options like if (input file), of (output file), and bs (block size).

```
2s [29] !dd

78-09-09
0+1 records in
0+0 records out
0 bytes copied, 22.1832 s, 0.0 kB/s
^C
```

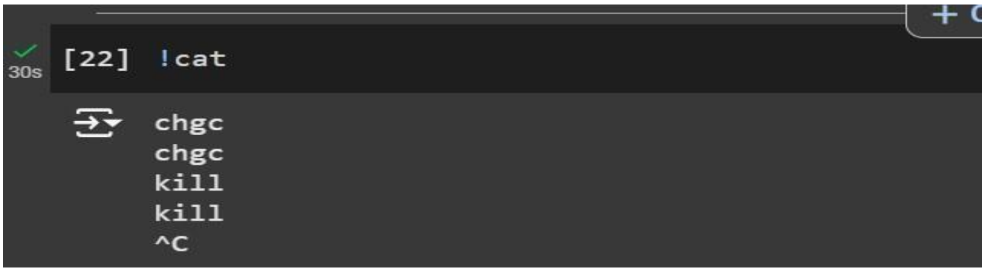
## 8. !echo

- Displays Text to the Console: The !echo command is primarily used to print a string or text to the output. For example, !echo Hello, World! will output "Hello, World!" in the cell. Can Be Used for Writing to Files: You can also use !echo to write output to a file by redirecting the output using the > operator. For example, !echo Hello > file.txt will write "Hello" into file.txt.




## 9. !cat

- Displays File Content: Running !cat filename will print the entire content of the specified file to the output, allowing you to view its contents directly.
- Can Be Combined with Other Commands: You can use !cat with other commands like | (pipe) to pass the output to further commands, such as searching with grep or paging through the content with less.




## 10.!clear

- Clears the Output: It removes all previous outputs in the terminal or console, providing a clean slate for new commands and results.
- Useful for Readability: It helps in maintaining a clutter-free workspace when running multiple commands in a notebook.

## 11. !whoami


- Displays the Current User: Running !whoami will output the username of the person or service currently running the Colab environment (usually root in Colab's case).

- Useful for System Information: It can help identify which user is executing commands in the environment, especially in shared or multi-user systems.


## 12. !service

- System Service Management (in Linux): The service command is typically used in Linux systems to manage system services, such as starting, stopping, or restarting services like web servers or databases. For example, !service apache2 restart would restart the Apache web server on a Linux machine.
- Not Fully Functional in Colab: Since Colab runs in a virtualized environment and doesn't provide direct access to system-level service management, commands like !service may not work or could return an error.


## 13. !mount

- Mounts External Drives or Storage: Typically, !mount is used in Linux systems to mount external devices, drives, or partitions. In Colab, it can be used to mount Google Drive to access its files within the Colab environment.
- Mount Google Drive: To mount Google Drive, Colab provides a specific method using drive.mount('/content/drive'). For example, running !mount alone will not function, but !mount can be used to display all mounted file systems.

```
!mount
```

```
overlay on / type overlay (rw,relatime,lowerdir=/var/lib/docker/overlay2/l/IVRTJAQXSADGI3KLQUJ74XI2KS:/v
proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
tmpfs on /dev type tmpfs (rw,nosuid,size=65536k,mode=755)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,gid=5,mode=620,ptmxmode=666)
sysfs on /sys type sysfs (ro,nosuid,nodev,noexec,relatime)
cgroup on /sys/fs/cgroup type cgroup2 (ro,nosuid,nodev,noexec,relatime,nsdelegate,memory_recursiveprot)
mqueue on /dev/mqueue type mqueue (rw,nosuid,nodev,noexec,relatime)
shm on /dev/shm type tmpfs (rw,nosuid,nodev,noexec,relatime,size=5989376k)
/dev/dm-0 on /usr/sbin/docker-init type ext2 (ro,relatime)
tmpfs on /tmp/colab_runtime.sock type tmpfs (rw,nosuid,nodev,noexec)
tmpfs on /var/colab type tmpfs (rw,nosuid,nodev,noexec)
/dev/sda1 on /etc/resolv.conf type ext4 (rw,nosuid,nodev,relatime,commit=30)
/dev/sda1 on /etc/hostname type ext4 (rw,nosuid,nodev,relatime,commit=30)
/dev/sda1 on /etc/hosts type ext4 (rw,nosuid,nodev,relatime,commit=30)
cgroup2 on /var/colab/cgroup/jupyter-children type cgroup2 (ro,relatime,nsdelegate,memory_recursiveprot)
```

## 14. !ps

- When executed, !ps displays a list of processes currently running on the system, including process IDs (PID), the terminal from which they were launched
- Can Be Extended with Options: You can use options to get more detailed output, such as !ps aux to display all processes running on the system, or !ps -ef for additional details about the processes.

```
!ps
```

```
  PID TTY          TIME CMD
    1 ?        00:00:00 docker-init
    7 ?        00:00:07 node
   22 ?        00:00:02 oom_monitor.sh
   24 ?        00:00:00 run.sh
   25 ?        00:00:01 kernel_manager_
   28 ?        00:00:00 tail
   39 ?        00:00:00 tail
   75 ?        00:00:17 python3 <defunct>
   76 ?        00:00:00 colab-fileshim.
   93 ?        00:00:08 jupyter-noteboo
   94 ?        00:00:03 dap_multiplexer
  391 ?        00:00:25 python3
  454 ?        00:00:06 python3
13447 ?        00:00:00 language_servic
13453 ?        00:00:30 node
14345 ?        00:00:00 sleep
14346 ?        00:00:00 ps
```

## 15. !unzip

- Extracts ZIP Files: Running !unzip filename.zip will extract the contents of the specified ZIP file into the current directory or a specified location.
- Can Be Customized: You can use various options with !unzip to control the extraction process. For example, !unzip -d /path/to/directory filename.zip will extract the contents into a specified directory.

```
!unzip
```

```
UnZip 6.00 of 20 April 2009, by Debian. Original by Info-ZIP.

Usage: unzip [-Z] [-opts[modifiers]] file[.zip] [list] [-x xlist] [-d exdir]
  Default action is to extract files in list, except those in xlist, to exdir;
  file[.zip] may be a wildcard.  -Z => ZipInfo mode ("unzip -Z" for usage).

  -p  extract files to pipe, no messages     -l  list files (short format)
  -f  freshen existing files, create none    -t  test compressed archive data
  -u  update files, create if necessary      -z  display archive comment only
  -v  list verbosely/show version info       -T  timestamp archive to latest
  -x  exclude files that follow (in xlist)   -d  extract files into exdir
modifiers:
  -n  never overwrite existing files         -q  quiet mode (-qq => quieter)
  -o  overwrite files WITHOUT prompting      -a  auto-convert any text files
  -j  junk paths (do not make directories)   -aa treat ALL files as text
  -U  use escapes for all non-ASCII Unicode  -UU ignore any Unicode fields
  -C  match filenames case-insensitively     -L  make (some) names lowercase
```

0s completed at 10:56 AM

## 16. !ssh

- Initiates Secure Shell Connections: In standard environments, !ssh username@hostname allows you to log into a remote machine via SSH to run commands and manage files.
- Limited Use in Colab: Since Colab is sandboxed and doesn't allow direct SSH access, using !ssh will likely result in an error unless specific services are set up for secure connections, which are outside the usual Colab functionality.

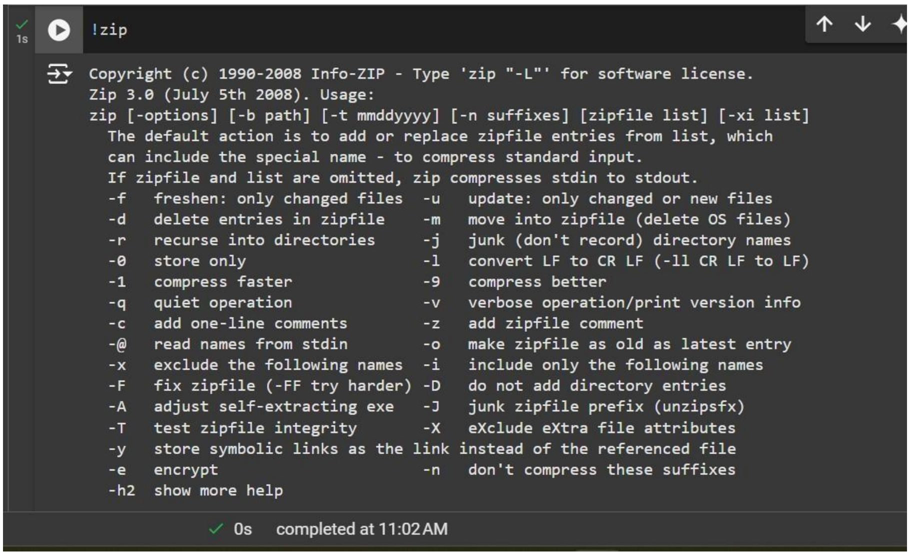
```
!ssh
```

```
usage: ssh [-46AaCfGgKkMNnqsTtVvXxYy] [-B bind_interface]
           [-b bind_address] [-c cipher_spec] [-D [bind_address:]port]
           [-E log_file] [-e escape_char] [-F configfile] [-I pkcs11]
           [-i identity_file] [-J [user@]host[:port]] [-L address]
           [-l login_name] [-m mac_spec] [-O ctl_cmd] [-o option] [-p port]
           [-Q query_option] [-R address] [-S ctl_path] [-W host:port]
           [-w local_tun[:remote_tun]] destination [command [argument ...]]
```

## 17. !zip

- Creates ZIP Archives: Running !zip archive_name.zip file1 file2 will create a ZIP file named archive_name.zip containing file1 and file2 (you can specify more files or directories as needed).

- Can Include Directories: You can use the -r option to recursively include all files from a directory in the ZIP archive. For example, !zip -r archive_name.zip directory_name will compress the entire directory and its contents.



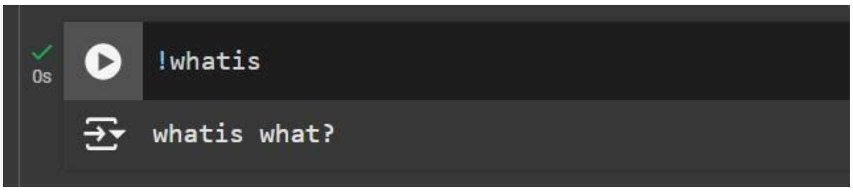
```
!zip

Copyright (c) 1990-2008 Info-ZIP - Type 'zip "-L"' for software license.
Zip 3.0 (July 5th 2008). Usage:
zip [-options] [-b path] [-t mmddyyyy] [-n suffixes] [zipfile list] [-xi list]
  The default action is to add or replace zipfile entries from list, which
  can include the special name - to compress standard input.
  If zipfile and list are omitted, zip compresses stdin to stdout.
  -f   freshen: only changed files  -u   update: only changed or new files
  -d   delete entries in zipfile    -m   move into zipfile (delete OS files)
  -r   recurse into directories     -j   junk (don't record) directory names
  -0   store only                   -l   convert LF to CR LF (-ll CR LF to LF)
  -1   compress faster              -9   compress better
  -q   quiet operation              -v   verbose operation/print version info
  -c   add one-line comments        -z   add zipfile comment
  -@   read names from stdin        -o   make zipfile as old as latest entry
  -x   exclude the following names  -i   include only the following names
  -F   fix zipfile (-FF try harder) -D   do not add directory entries
  -A   adjust self-extracting exe   -J   junk zipfile prefix (unzipsfx)
  -T   test zipfile integrity       -X   eXclude eXtra file attributes
  -y   store symbolic links as the link instead of the referenced file
  -e   encrypt                      -n   don't compress these suffixes
  -h2  show more help

0s   completed at 11:02 AM
```

## 18.!whatis

- Displays Short Description of Commands: In standard Linux environments, whatis provides a one-line description of a command. For example, whatis ls would output a brief description of the ls command.
- Not Available in Colab: Since the whatis command relies on a manual database that isn't available in Colab, attempting to use it will not yield any result.



```
!whatis

whatis what?
```

## 19. !apt

- Installs and Manages Packages: You can use !apt to install various software packages that are not pre-installed in the Colab environment. For example, !apt-get install packagename installs the specified package.

- Package Management: It can also be used for other package management tasks like updating the package list (!apt update), upgrading installed packages (!apt upgrade), or cleaning up unnecessary packages (!apt autoremove).

```
!apt

apt 2.4.13 (amd64)
Usage: apt [options] command

apt is a commandline package manager and provides commands for
searching and managing as well as querying information about packages.
It provides the same functionality as the specialized APT tools,
like apt-get and apt-cache, but enables options more suitable for
interactive use by default.

Most used commands:
  list - list packages based on package names
  search - search in package descriptions
  show - show package details
  install - install packages
  reinstall - reinstall packages
  remove - remove packages
  autoremove - Remove automatically all unused packages

0s completed at 11:05 AM
```

## 20. !sort

- Sorts Lines in Ascending Order: By default, !sort sorts the lines of input in lexicographical (alphabetical) order. For example, running !sort on a file or input will arrange the lines from lowest to highest.
- Can Sort in Reverse: You can use options like -r to sort in reverse order (descending).

For example, !sort -r will sort the input from highest to lowest.

```
!sort

9,8,7
7,8,9
^C
```

## Conclusion:

In this demonstration, we explored a range of basic Linux commands that can be used in

Google Colab to interact with the system, manage files, and retrieve information. These commands provide functionality for managing files (e.g., ls, cat, zip), monitoring system performance (e.g., ps, df), and performing general tasks like installing software (apt-get) and viewing system information (whoami, date). While Google Colab's environment doesn't support all Linux commands (like those involving service management or SSH), many essential file and process management commands are fully functional and can enhance your workflow in Colab.